

MODULE 05

Hip MRI & THA: AVN Staging, FAI & Arthroplasty Complications

Ficat-Arlet · ARCO · Alpha Angle · LCEA · FAI · Vancouver · ALTR · Brooker

1.0 AMA PRA Cat 1

Hip

CME Article Review

LEARNING OBJECTIVES

1. Apply Ficat-Arlet and ARCO classifications to femoral head avascular necrosis (AVN) on MRI and radiograph.
2. Calculate and interpret alpha angle, lateral center-edge angle (LCEA), and femoral anteversion on hip MRI.
3. Classify femoroacetabular impingement (FAI) subtypes — cam, pincer, mixed — and their characteristic MRI features.
4. Identify and grade THA complications including ALTR (Anderson classification), trunnionosis, particle disease/osteolysis, and periprosthetic fracture (Vancouver classification).
5. Apply the Brooker classification to heterotopic ossification following THA.
6. Recognize the crescent sign as the critical imaging juncture for joint-preserving surgery in AVN.

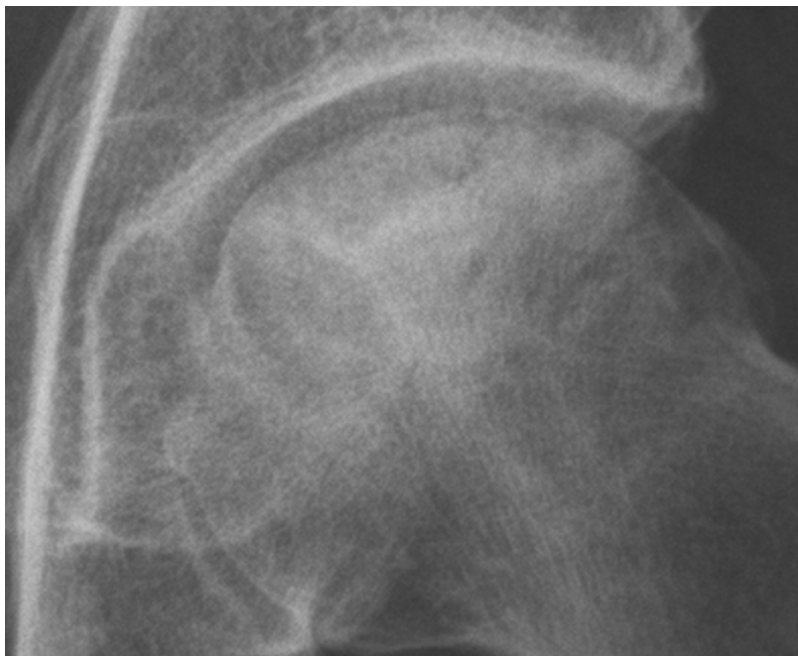
This module is produced by the LucidMSK Editorial Board for educational purposes only. Upon completion of all content and a post-test score $\geq 70\%$, participants may claim the stated 1.0 AMA PRA Category 1 Credit(s)[™]. No commercial support was received.

1. Avascular Necrosis (AVN) — Ficat-Arlet & ARCO Classifications

Avascular necrosis (AVN), also called osteonecrosis, of the femoral head is graded to guide management decisions, since the window for joint-preserving surgery closes once subchondral collapse occurs. MRI is the most sensitive modality, detecting marrow signal changes well before any radiographic abnormality is visible. The Ficat-Arlet system was originally based on radiographic findings; the ARCO (Association Research Circulation Osseous) system incorporates MRI from the outset and adds quantitative lesion-size subtyping, making it more comprehensive for modern practice.

Ficat-Arlet Classification — Femoral Head AVN

Stage	Radiograph	MRI Finding	Management
I	Normal	Double-line sign on T2; band-like T1 hypointensity	Core decompression ± bone marrow concentrate
IIA	Sclerosis or cysts; no crescent sign	Established marrow signal change, no collapse	Core decompression ± grafting
IIB	Crescent sign — subchondral fracture	Curvilinear T2 fluid signal beneath the articular cortex	Core decompression; vascularized fibular grafting; impending collapse
III	Femoral head collapse; acetabulum still normal	Flattening of the articular surface evident	Hip resurfacing (younger patients); THA (older patients)
IV	Collapse plus secondary degenerative (osteoarthritic) change	Established arthrosis changes	Total hip arthroplasty required



Radiograph showing subtle subchondral lucency and early collapse of the femoral head, consistent with Ficat-Arlet Stage III / ARCO Stage 3 AVN.

ARCO Classification — Femoral Head AVN

Stage	Findings	Lesion Size (ARCO Key Addition)
0	Biopsy-proven necrosis; all imaging (radiograph, CT, bone scan, MRI) normal	N/A
1	Normal radiograph; abnormal MRI/bone scan; band-like T1 hypointensity	A < 15%, B 15–30%, C > 30% of the femoral head
2	Radiograph shows mottled sclerosis, lysis, or osteopenia; no crescent sign or collapse	Subtyped A/B/C by percent head involvement on MRI
3	Subchondral collapse (crescent sign or flattening); acetabulum still normal	A < 2 mm depression; B 2–4 mm; C > 4 mm
4	Established osteoarthritis: joint space narrowing; acetabular involvement	N/A



Radiograph demonstrating femoral head flattening with collapse and secondary joint space narrowing, consistent with Ficat-Arlet Stage IV / ARCO Stage 4 AVN with established osteoarthritis.

PITFALL

The crescent sign — a curvilinear subchondral lucency on radiograph or a corresponding band of T2-hyperintense fluid signal on MRI — defines ARCO Stage 3 / Ficat-Arlet Stage IIB. It indicates a subchondral fracture and impending collapse, and represents the last opportunity for joint-preserving surgery before the femoral head deforms irreversibly. Its presence or absence should be stated explicitly in every AVN report.

REFERENCES

- Ficat RP. Idiopathic bone necrosis of the femoral head: early diagnosis and treatment. *J Bone Joint Surg Br* 1985.
- Mont MA et al. Current concepts review: osteonecrosis of the femoral head. *J Bone Joint Surg Am* 2010.

2. Femoroacetabular Impingement (FAI) — MRI Measurements

Femoroacetabular impingement results from abnormal contact between the proximal femur and acetabular rim, leading to labral and chondral damage over time. FAI is divided into cam, pincer, and mixed subtypes based on which side of the joint drives the abnormal morphology, and a specific set of angular measurements on radial MRI sequences is used to confirm and characterize each subtype.

FAI Subtypes

Type	Morphology	Key Measurement	Cartilage Damage Pattern
Cam	Aspherical femoral head; "pistol-grip" deformity; reduced head-neck offset	Alpha angle > 55°	Superior/anterosuperior acetabular cartilage ("outside-in" delamination)
Pincer	Acetabular overcoverage; acetabular retroversion	LCEA > 40°; positive crossover sign	Posterior labrum and cartilage (contrecoup damage pattern)
Mixed	Combined cam and pincer morphology	Combined alpha angle and LCEA abnormalities	Combined pattern; the most common type encountered clinically

Key Hip MRI Measurements for FAI

Measurement	Normal	Abnormal Threshold	Significance
Alpha angle (radial oblique MRI)	< 55°	> 55° = cam morphology	Head-neck asphericity driving anterosuperior impingement
Lateral center-edge angle (LCEA, Wiberg)	25–40°	< 25° = dysplasia; > 40° = overcoverage (pincer)	Degree of lateral acetabular coverage of the femoral head
Femoral anteversion	10–25°	< 10° or > 35° = abnormal	Rotational profile of the femur; abnormal values predispose to instability or impingement
Tönnis angle (acetabular roof angle)	< 10°	> 10° = undercoverage	Acetabular roof obliquity; elevated values suggest dysplasia

PEARL

An alpha angle measured at 68° at the anterosuperior head-neck junction on radial oblique MRI is clearly elevated above the 55° cam threshold and should be reported as cam morphology, with correlation to the pattern of cartilage and labral damage seen at the corresponding clock-face position.

REFERENCES

- Notzli HP et al. The contour of the femoral head-neck junction as a predictor for the risk of anterior impingement. *J Bone Joint Surg Br* 2002.
- Wiberg G. Studies on dysplastic acetabula and congenital subluxation of the hip joint. *Acta Chir Scand* 1939.

3. THA Complications — ALTR, Trunnionosis, Vancouver & Brooker

Total hip arthroplasty (THA) surveillance imaging requires a systematic checklist approach, since several distinct failure mechanisms can coexist and each carries a different management pathway: adverse local tissue reaction (ALTR, sometimes called pseudotumor), trunnionosis (corrosion at the femoral head-neck taper junction), particle disease with osteolysis, periprosthetic fracture, and heterotopic ossification.

Anderson ALTR Classification — Metal Artifact Reduction Sequence (MARS) MRI

Type	Description	Risk Level / Management
1 — Aseptic lymphocytic synovitis	Synovial thickening with joint fluid; no discrete mass	Low — interval monitoring
2A — Solid pseudotumor	Solid, well-defined periarticular mass	Moderate — surgical if symptomatic or enlarging
2B — Cystic pseudotumor	Cystic or complex fluid collection with a thin wall and no solid component	Moderate — surgical if large or symptomatic
3 — Mixed	Combined solid and cystic components with aggressive features	High — early revision recommended
4 — Destructive	Bone destruction with muscle or tendon involvement	Critical — urgent revision

Trunnionosis refers to progressive mechanical and chemical corrosion at the femoral head-neck taper junction, most often encountered with metal-on-metal (MoM) bearings and large-head metal-on-polyethylene (MoP) constructs. If unrecognized and unchecked, trunnionosis can progress to catastrophic head-neck dissociation — among the most severe THA complications. Findings on imaging can be subtle: look for metallic debris in the periarticular soft tissues and any asymmetry in femoral head position relative to the trunnion.

Vancouver Classification — Periprosthetic Femoral Fracture (THA)

Type	Location	Stem Stability	Surgical Strategy
AG	Greater trochanter	N/A	ORIF if displaced > 2.5 cm (abductor mechanism at risk)
AL	Lesser trochanter / proximal calcar	Usually stable	Conservative management if < 2 cm displacement and a stable stem
B1	At or around the stem tip	Stable	ORIF with plate and cerclage; the stem is retained
B2	At or around the stem tip	Unstable (loose)	Revision to a long-stem prosthesis
B3	At or around the stem tip	Unstable with poor bone stock	Revision with allograft or proximal femoral replacement
C	Distal to the stem tip	N/A (stem uninvolved)	Independent ORIF; the stem is retained regardless of any incidental loosening

PITFALL

The B1 versus B2 distinction is the single most consequential decision point in this classification. Both occur at or around the stem tip, but stem stability determines the entire surgical approach — ORIF with the stem retained for B1, versus revision arthroplasty for B2. Thin-cut CT can help assess stem fixation by identifying subsidence or circumferential lucent lines, both of which indicate B2-type loosening.

Brooker Classification — Heterotopic Ossification (Post-THA)

Grade	Finding	Clinical Impact
I	Islands of bone within the soft tissues	Minimal
II	Bone spurs from the pelvis and/or proximal femur, with ≥ 1 cm gap between opposing surfaces	Mild; monitor
III	Bone spurs with < 1 cm gap between opposing surfaces	Moderate; range-of-motion restriction
IV	Bony ankylosis of the hip	Severe; surgical excision required after bone maturity

CLINICAL TIP

Grade IV heterotopic ossification should not be excised prematurely. Excision is delayed until the bone has matured — confirmed by bone scan quiescence, typically at 12–18 months post-operatively — since earlier excision carries a substantial risk of recurrence. Indomethacin or local radiation prophylaxis is often used for a planned contralateral THA in patients with a history of significant heterotopic ossification.

REFERENCES

- Duncan CP, Masri BA. *Fractures of the femur after hip replacement. Instr Course Lect* 1995.
- Anderson H et al. *Adverse local tissue reaction (ALTR): classification with MARS MRI. J Arthroplasty* 2017.
- Brooker AF et al. *Ectopic ossification following total hip replacement: incidence and a method of classification. J Bone Joint Surg Am* 1973.